

# Not transparent and incomprehensible: A qualitative user study of an AI-empowered financial advisory system

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## ABSTRACT

AI-empowered and algorithm-driven automated financial advisory systems, also known as Robo-advisors, have been rapidly implemented by service providers and customers in financial service markets. Yet, few empirical studies investigate customers' experience interacting with fully functional Robo-advisors in real-life scenarios. Also, it is still unknown how the design of the automated system can affect customers' perception and adoption of this new technology. To mitigate these gaps, 24 participants with different levels of experience and understanding of financial investment were asked to use a Robo-advisor from a retail bank and perform the tasks. By conducting observations and retrospective post-test interviews, we find that participants do not fully perceive the social aspects supposed to be provided by Robo-advisors. The overarching problems are, among others, a lack of transparency and incomprehensible information. This results in distrust of the results generated by this system, which negatively affects customers' adoption of the investment advice provided by the Robo-advisor. The potential of interactive data visualization is also detected. This work contributes to the understanding of customers regarding their perception and adoption based on their use of a functional Robo-advisor and proposes design takeaways for transparent and comprehensible automated advisory systems in financial service contexts.

## 1. Introduction

AI-empowered and algorithm-driven automated systems have been rapidly implemented and are commonly used in people's everyday life. This type of technology enables the automation of services by eliminating human intervention; and has been claimed to provide disruptive potential and is widely used across service sectors and industries, including managerial decision-making (Jarrahi, 2018; Shrestha, Ben-Menahem, & Von Krogh, 2019), customized recommendations (Jameson, Konstan, & Riedl, 2002; Namjun, Hosoo, Sangman, & HWANG, 2019), medical care (Cai, Winter, Steiner, Wilcox, & Terry, 2019; Holzinger, Biemann, Patichis, & Kell, 2017; Yang, Steinfeld, & Zimmerman, 2019), and automobiles (Azaria, Rosenfeld, Kraus, Goldman, & Tsimhoni, 2015). In the personal finance context, a type of algorithm-driven automated financial advisory service on the mobile terminal, also known as "Robo-advisor" or "Robo-advisory" in business management and financial communities, has become popular among customers of retail banks and FinTech companies, especially during the COVID-19 when people have become more open for digital and automated service (Ben-David & Sade, 2020; Gan, Khan, & Liew, 2021; Guo & Polak, 2021; Insider Intelligence, 2021).

The emergence and development of Robo-advisors (RAs) were initially driven by financial service providers (e.g., retail banks) to replace human advisors' services to cut labor costs and transfer traditional services from offline to online (Fisch, Labouré, & Turner, 2019). Also, service providers want to attract more customers who are different from those in conventional advisory services by promoting RAs' low cost, flexibility, accessibility 24/7, and the same service quality as a human advisor. Robo-advisors have shown their potential for mitigating biases and conflicts of interest in human advisory encounters (Bolton, Freixas, & Shapiro, 2007; Burke, Hung, Clift, Garber, & Yoong, 2015; Faloon & Scherer, 2017) and have proven their commercial values in markets. UP to January 2022, two independent leading RA companies in the world reported double-digit increases in account openings during the pandemic (Insider Intelligence, 2021).

Despite RAs' rapid growth, compared with the conventional advisory service, the perception of RAs is unclear, and their adoption is still low among customers in the financial service market (Brown, 2017; Erin EI, 2020; Jung, Dorner, Weinhardt, & Puzmaz, 2018). In Information Systems and Business Studies, research mainly focuses on customers' perception of RAs in general and their adoption of the final

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recommendation (i.e., investment portfolio) generated by RAs. In most cases, the adoption in these articles indicates customers' intention to use RAs. Many influencing factors were reported from different aspects, such as user demographics, their investment intention, trust, perceived usefulness, background knowledge, and social media (Ben-David & Sade, 2020; Gan et al., 2021; Hildebrand & Bergner, 2021; Hohenberger, Lee, & Coughlin, 2019; Lourenço, Dellaert, & Donkers, 2020; Seiler & Fanenbruck, 2021; Wu & Gao, 2021). Most of these studies analyzed customer survey data and developed models based on TAM (Technology Acceptance Model) or UTAUT (Unified Theory of Acceptance and Use of Technology) theories. Their respondents had no experience using Robo-advisors, and the users were limited to certain groups, such as active investors. Understanding the motivations and existing obstacles to using and adopting RAs can provide insights for service providers to implement marketing strategies. However, these studies have limited contributions to improving existing system design because the role of a Robo-advisory system is still unknown in terms of user perception and adoption. Thus, empirical studies should switch to a more user-centered approach to data based on customers' experience of using a RA. And more attention should be paid to how the system can influence users' perception and adoption.

Transparency and trust are complex and particularly significant topics when AI-empowered systems are contextualized in customers' financial decisions. Financial decisions can be high-stake behaviors and customers face the possibility to lose their money. Customers in financial services, unlike users in other automated systems, tend to be "vulnerable" (Mogaji, Soetan, & Kieu, 2021). Any unfair or untrustworthy perception will adversely and negatively affect customers' adoption. Customers' understanding of the information provided by RAs is also essential. Due to the lack of human experts' assistance or explanation, customers highly depend on their own understating when using RAs, in which, most information is in written form or data visualization (Salo & Haapio, 2017). Design research in the internet-based era came up with solutions as collaborative interfaces or platforms that enhance perceived transparency and comprehensibility (Kilic, Heinrich, & Schwabe, 2015; Nueesch, Puschmann, & Alt, 2014, 2016; Nussbaumer, Matter, Reto à; Nussbaumer & Matter, 2011; Porta, et al., 2012). However, it is difficult for these design solutions to adapt to the AI-empowered system. Without human intervention, the "black-box" nature of the algorithm embedded in this system makes it difficult for users to understand why and how this decision (i.e., recommended investment portfolio) is made and claimed to be "optimized" for them. This might lead to distrust and low adoption even if the AI's decision is accurate (Dikmen & Burns, 2022). Two approaches are detected by researchers in Human-computer/robot Interaction (HCI/HRI). One of them is the anthropomorphic design which aims to compensate for the lack of human advisors in the automated process (Adam, Toutaoui, Pfeuffer, & Hinz, 2020; Deng & Chau, 2021; Hildebrand & Bergner, 2021; Morana, Gnewuch, Jung, & Granig, 2020). But the requirement of human-like design has not been theoretically problematized in empirical studies. Even though those studies have proved that the perceived social presence can delegate more trust, these studies have not shown whether a human-like robot is useful for achieving customers' goals in financial service contexts. The other approach is transparent and responsible AI systems in financial services (Anshari, Almunawar, Masri, & Hrdy, 2021; Stefanel & Goyal, 2019; Zheng, Zhu, Li, Chen, & Tan, 2019). Yet, existing research rarely illustrates which type of transparency is lacking and what is needed based on users' lived experiences and how to incorporate users' demands into existing systems. Moreover, RAs are multi-stakeholder autonomous systems (Sonboli, Smith, Cabral Berenfus, Burke, & Fiesler, 2021), in which the customer's profits are integrated with providers' business goals and financial regulations. All these factors make it a complex but urgent issue to figure out the relationship between RAs' system design and customers' perception and adoption of RAs.

The research gaps identified in previous literature are: (1) few empirical studies investigate customers' experience interacting with fully

functional RAs in real-life scenarios; (2) the relationship between the design of the automated system and customers' perception and adoption of RAs is still unknown. This study aims to understand customers' experience of using a RA, particularly to shed light on how the system design can affect their perception and adoption of RAs. Perception focuses on customers' holistic experience and understanding of the RA system and service based on their interaction throughout all aspects of the customer's journal, from onboarding to the final generated portfolio. Adoption addresses their intention and willingness to use RAs in the future after they review the final recommendation generated by RAs based on their real-life situation (e.g., personal income, risk appetite, loans and savings, etc.). The 24 participants with different levels of investment experience and financial understanding joined the lab experiment, and their demographics were also collected and considered in the analysis. This paper discusses fully automated systems on the mobile end, so a hybrid system is not within the scope. The user test approaches from HCI include pre-test surveys, assigning scenarios and tasks, observing performance, and collecting feedback from retrospective post-test interviews. The research questions include: (1) how are customers able to interact with a retail bank RA system; and (2) how does this experience affect their perception and adoption of this RA system?

This article briefly introduces the RA system and its implementation in financial service sectors. First, we present an interdisciplinary literature review of the state-of-art research that is relevant. Then, the empirical user study is introduced regarding the test procedure, participant recruitment, data collection, and analysis. Because most RAs in financial markets have similar interfaces and features. The selected RA in the empirical study is a representative RA application that enjoys a large number of users and is produced by a well-known financial group in Scandinavia countries. Its name is anonymous. Last, findings are presented and discussed. This work finds the lag in customers' perception of RA compared with its business ambitions. Also, the perceived non-transparency and incomprehensible information in the RA system cause distrust, and those experiences negatively affect their adoption of the portfolio generated by RA. However, the visualization tool is promising to fill these gaps.

This work extends the knowledge of system design in customers' perception and adoption of AI-empowered decisions in financial services by contributing empirical evidence and proving the user test as an effective method. Also, it elicits the existing problems that can be improved in the design of AI-empowered automated systems. Most importantly, it articulates the demand from customers for a RA system, improving transparency and comprehension. This compensates for prior RA empirical studies by asking participants to use a well-functional retail bank RA and view the result generated by AI.

## 2. Literature review

Starting with an introduction to RAs' implementation, this part summarizes existing research regarding RAs from interdisciplinary perspectives (e.g., information systems, service markets, design science) and their challenges that can benefit from this study. This section also includes the key concepts recurring in this study, such as transparency and trust.

### 2.1. The implementation of RAs in financial service

RAs use algorithms to provide customers with financial advisory services and support their investment decisions. This implementation is standardized in the markets but might be slightly different according to laws and regulations. Normally, these systems include four stages: (i) personal data collection (customer's savings, incomes, and preferences), (ii) risk assessment (using a standardized survey to categorize customers into different risk-apatite groups), (iii) wealth management advice (providing an investment portfolio based on collected data and algorithms), and (iv) continuous investment maintenance (Fisch et al., 2019;

Jung, Dorner, Glaser, & Morana, 2018; Jung, Glaser, & Köpplin, 2019). Besides, some additional phases are implemented according to national regulations and governances, for example, a basic financial literacy test or a sustainable investment alternative (European Commission, 2021). Some can even harvest tax losses (Berger, 2015). Usually, the complete RA workflow is **Configuration** (i & ii), **Matching and Customization** (iii), and **Maintenance and Rebalancing** (iv) (Jung, Dorner, Glaser, & Morana, 2018; Jung et al., 2019). In the Configuration Phase, by filling in questionnaires or answering specific questions, users' personal information, financial situation, risk tolerance, and preferences are collected for customer profiling and assessment. The RAs are supposed to identify users' requirements and aims of financial investment. Then, users are automatically classified into different groups according to individual profiles by the system. In Matching & Customization, an undisclosed algorithm is deployed to match a customer's profile to a portfolio recommendation (i.e., a collection of different types of financial investment products like stocks, bonds, or exchange-traded funds [ETFs]). Like other algorithm-based systems, this step is a "black box" (Litterscheidt & Streich, 2020). Data visualization is commonly used in this step to convey investment planning and the estimated performance of the recommended portfolio. In Maintenance and Rebalancing, the system automatically monitors and rebalances the portfolio to meet the optimal model.

Meanwhile, its performance and adjustment will be regularly communicated and informed to the users. Even though users can reach customer service anytime they need support, no human intervention is expected during the whole process; therefore, the "Robo-" here indicates the fully automated self-service. RA is initially used in business and financial communities, then extended to IS and HCI/HRI research.

## 2.2. Studies to understand RA customers

Because RAs are still emerging phenomena, it is of great value for developers and service providers to understand the user demographics, motivations, attitudes, and expectations. Studies revolving around users' perception and adoption are mainstream objectives in Information Systems and Business Studies in Financial Market. These user studies collected survey data from users to establish trust and adoption models of RAs (e.g., unified theory of acceptance and use of technology) across temporal (e.g., at COVID-19 crisis) and regional scopes (e.g., Germany and Malaysia.) (Lourenço et al., 2020; Seiler & Fanenbruck, 2021; Todd & Seay, 2020; Wu & Gao, 2021). Their models help to explain the influential factors that lead users either to adopt the RA's final recommendation or invest money by accepting this service. In those models, some factors regarding customers' experience of RAs were mentioned but have not been deeply discussed by connecting them with RA system design. Specifically, those factors, including the perceived usefulness and performance expectations, were collected from survey data and were not based on their real experience using RAs. To compensate for this, a user-centered design study was conducted for risk-averse and low-budget customers (Jung, Dorner, Weinhardt, & Puszma, 2018). Their work contributes principles to RA design: ease of interaction, work efficiency, information processing, cognitive load, and advisory transparency. These principles, especially the last two regarding the user's cognitive load and transparency, are essential design takeaways for RA system design. However, these design principles were limited by focusing on one specific customer group, and because the tested system was still in development and iteration, the final recommendation generated by the RA was not discussed in this paper.

## 2.3. Anthropomorphic robot to delegate trust

Financial and investment decisions have traditionally and historically been about trust. Trust can be defined as "the willingness to take actions of another entity based on the expectation that the other will perform a particular action, regardless of the ability to monitor or control that other part" (Bedué & Fritzsche, 2022, p. 533). Thus, in this case, trust is about if

RAs can provide the expected benefits that customers feel entitled to. To maximize trust, human advisors have been involved in this process in the conventional financial service encounter, while they have been increasingly augmented with AI-empowered technology over time. In traditional service encounters, the social interaction and relationship between human advisors and customers significantly affect customers' perception of advisor credibility and willingness to follow their advice (Söderberg, 2013). With the AI replacement, the higher trust towards the Robo advisor, the more likely consumers will adopt its recommendation (Bruckes, Westmattmann, Oldeweme, & Schewe, 2019). Thus, the social and anthropomorphic design could be alternative solutions to delegate trust to compensate for the lack of human intervention in the RAs (Bruckes et al., 2019). These studies usually articulate the perceived social presence as anthropomorphism in RAs. For example, researchers investigated verbal and visual cues (Adam et al., 2020), social cues in conversational chat (Day, Lin, & Chen, 2018; Hildebrand & Bergner, 2021), visually human-like design (Deng & Chau, 2021), and varying degrees of a human-like chatbot (Morana et al., 2020). These have proven the possible influence of perceived social presence towards a higher trust and adoption of RAs and the final recommendation. However, the human-like design in business implementation and financial services is not a one-size-fits-all solution because other factors dynamically influence trust (Cheng et al., 2019). Also, trust delegated by human-likeness robots is vulnerable to being disrupted by factors such as the unsatisfied experience of using chatbots or a lack of understanding (Dikmen & Burns, 2022). An empirical study shows that compared to responsiveness, customization, and such traits, likability is not significant to customers' intention to use chatbots in financial services (Lee & Park, 2019).

## 2.4. Information system in fin-tech

RAs are automated decision-making systems, fundamentally disruptive compared with other digital financial services provided by banks or other retail financial institutions. Information systems are also applied in human-advisory encounters, namely, a collaborative platform as a tool for human advisors to advise customers. Prior research addressing IT-support (web-based) systems in digital financial services has detected that transparency and information symmetry are important principles (Kilic et al., 2015; Nueesch et al., 2014, 2016; Nussbaumer, Matter, Reto à, Nussbaumer & Matter, 2011; Nussbaumer, Matter, & Schwabe, 2012; Porta, et al., 2012). These studies also indicate that the system design of digital advisory services is as important as the performance of the recommended financial products. Compared with these digital systems where human intervention still exists, an automated system is supposed to promise more transparency regarding cost and the advisory process (Tertilt & Scholz, 2018). However, these studies in financial service systems focused on web-based services rather than a fully-automated system, i.e., RA.

According to Turilli and Floridi (2009), transparency is the process of making information explicitly and openly available so that users can exploit the disclosing information to make decisions. In information systems and design science, the challenge of information transparency can be addressed by Explainable AI (XAI). According to Eschenbach (2021), transparency is necessary for trust and determining if AI is trustworthy. Based on this philosophy, XAI can be a lens for researchers to uncover users' acceptance and trust in AI (Shin, 2021b). XAI has been implemented and examined in varying settings [e.g., mental health (Wolf & Ringland, 2020), recommendation system (Shin, 2021a)], as well as in the RA context (Ben David, Resheff, & Tron, 2021). These studies discovered the potential of using XAI for designing a RA that builds trust and adoption, even though most of them were not contextualized in financial contexts.

A second challenge normally is non-expert users who lack the domain knowledge to comprehend and interpret the explanation information provided by this AI-empowered service. Empirical studies have shown

that the typical RA user is young with limited experience in the financial market (Fisch et al., 2019; Insider Intelligence Editors, 2021), low-risk, low-budget (Jung, Dorner, Weinhardt, & Pasmaz, 2018), and with limited financial literacy (Brenner & Meyll, 2020). This demonstrates that discussions around “comprehensible” RA systems are important because, more than others, non-expert customers need an expert system to help them make an investment decision. Prior research shows that domain knowledge can help non-expert customers with less financial understanding and investment experience interpret the explanation provided by the AI-empowered system (Dikmen & Burns, 2022). In RAs, domain knowledge usually indicates financial or investment understanding, and it is known as financial literacy when systematically measured by a standardized test. Data visualization has been proven to be an effective tool in supporting customers’ understanding and decision making (Eberhard, 2021; Perdana, Rob, & Rohde, 2018; Tang, Hess, Valacich, & Sweeney, 2014). However, it is still unknown how the financial understanding is delivered by existing RA systems. Also, the usage and effect of interactive data visualization in RAs remain to be investigated by empirical studies.

### 3. Methodology

This user test includes an introduction, pre-survey, scenario setting, task performance by using the RA application, and a post-test retrospective interview. The survey data, interview transcriptions, and screen recordings of participants’ interaction with the interface were analyzed accordingly.

#### 3.1. Participants

To complement the participant groups of prior empirical studies that focus on either specific user groups (Jung, Dorner, Weinhardt, & Pasmaz, 2018; Morana et al., 2020) or people who are actively in investment activities (Hildebrand & Bergner, 2021), here we aimed to recruit participants coming from more diverse demographic groups in terms of ages and different levels of financial understanding and investment experience. This choice also considers that the customers inclined to use RAs are changing over time (Fulk, Grable, & Kruger, 2018; Gan et al., 2021; Pradhan & Wang, 2020). This study does not aim to systematically compare differences between participant groups, but some patterns found between groups were observed and will be reported in the finding section.

Two participants performed the pilot test, and 24 participants with stable incomes by convenience sampling joined the formal user test. The participants in this study have never used the kind of automated advisory service investigated in this study before. All participants have stable incomes (MED. = €300 to € 400 per month). The participants either had experience with the financial investment before or were interested in trying this service to see the possibility of financial investment in the future. According to their self-reported financial knowledge and experience, they were divided into two groups: non-expert customers (P2, P3, P5, P6, P8, P10, P12, P13, P14, P16, P18, P19) and expert customers (P1, P4, P7, P9, P11, P15, P17, P20, P22, P23, P24). The division into two groups was not made to rigorously compare the groups but to identify patterns within different groups by analyzing the screen recordings. The professions of participants varied from researchers (including PhD students), engineers, data analysts, administrators, physicians, librarians, etc.

#### 3.2. Design of the user test

Inspired by the *retrospective think-aloud approach* (van den Haak, De Jong, & Jan Schellens, 2003), a combination of that method and a semi-structured post-test interview method was used in this study. This is a retrospective post-test interview. Participants were not asked to “think aloud” during the test but talked about their performance and intentions

by referring to the screen recording of their task performance during the interview. The reason for combining these methods was that the think-aloud protocol normally used in usability tests is somewhat unsuitable for this study. Because participants need a quiet and secure environment when inputting their financial data, such as loans, savings, and incomes, in real-life scenarios, they should concentrate more on dealing with complex information instead of speaking out about mental activities while doing the tasks.

Thus, in the formal user tests, the screen recording would be shown to participants as a reference and reminder when they, during the post-test interview, were asked about their actions or if they actively mentioned certain operations and thoughts about details in their interaction with the interface. Instead of evaluating the usability of the existing system, this user test primarily focused on understanding how users processed information underneath behaviors and their perceptions and behavioral intentions (i.e., perceptions) after interacting with a RA. This test aimed to detect existing concerns and problems through self-explanations, supported by the interview questions rather than the researchers’ observations.

#### 3.3. Procedure

All the user tests were conducted in a quiet room in a lab environment. The complete user test contained three sessions: (1) pre-survey, to collect participant’s demographic information, self-reported experience and knowledge in financial investment, and frequency of using technology and AI system; (2) performing three tasks by using the RA (video is recording at the same time); and (3) post-test retrospective interview, answering the questions in the interview, in parallel with describing and explaining their behaviors by referring to the video playback.

Before the user test, each participant was informed of the aims and process of this test; and was asked to sign the consent form regarding information usage and privacy protection.

##### 3.3.1. Tasks and scenarios

One task with scenarios was presented to the participants in a written format before and while they were using the RA: (1) think about a scenario in which you want to invest your savings for a long- or short-term target; (2) try this RA as one of your investment options and explore this automated service by yourself; (3) stop before the system asks for your final decision and keep in mind whether you want to use it or not in the future and if you want to adopt the final recommendation.

One researcher stayed in the lab as a moderator to offer help if the participant had any problems. This moderator did not interact with the participant during the task performance part of the test to simulate the automated self-service process and encourage participants to finish the service themselves. However, if the participant had problems that prevented them from proceeding with the test, this researcher would offer some assistance regarding operations to ensure that the test could proceed. This helps participants to complete the advisory process to have a holistic understanding of the RA service, starting from the onboarding experience until the final recommended portfolio.

##### 3.3.2. Post-test interview

The first author constructed the first draft of the interview protocol; then, it was reviewed by the co-authors specializing in socio-robotics and financial services. Some texts were reworded, and a few questions were added. The final interview protocol contains a list of questions encompassing three parts. The first part concerns participants’ prior investment channels and how they differ from the RA they experienced in the test. In this part, a semi-structured interview method was used. In the second retrospective interview part, the questions focused on participants’ experience when using and interacting with the RA. Here, the researcher re-played the video recording and asked participants for feedback regarding each step and the overall process. For example, any problems they encountered when using it, their expectations of the system, and



unmet needs. In the third part, questions were asked about what participants thought about the final recommendation that was customized according to their input. In the fourth part, participants were asked whether they thought a virtual client service or chatbot is more helpful and supportive in this type of product compared to a human advisor.

### 3.4. Apparatus

#### 3.4.1. Test hardware and software

In terms of the test hardware, researchers pre-installed the RA application in a test smartphone (iPhone 6s Plus). And the software is a selected RA by a retail bank.

The RA selected for this study was developed by a bank, one of Sweden's largest financial groups. The selected RA is also one of the earliest launched automated financial services, among other similar products. Using a RA service provided by a big bank can help to recruit participants much easier due to its trustworthiness and authority perceived by investors, and testing a mature application with accurate technological support can help to reduce the consumers' uncertainty caused by inaccurate algorithms that might affect the users' perception and trust of the RAs.

The system design of most RAs on the mobile end is rather simple and linear (see Fig. 1). The selected RA has the same usage flow. Users can navigate by pressing the “back” and “next” buttons, scrolling up and down to view the interface, inputting numbers and sliding the bar to change numbers when interacting with data visualizations. Additionally, supplementary documents, such as fact sheets of financial products, risk disclosure, and purchase agreement, can be reviewed as attachments by pressing the icon, which directs users to a PDF document.

#### 3.4.2. Documentation

Each interview audio was recorded on an iPad for future transcription and analysis. An iOS video recording software was used to record the participants' interaction with the interface during the user test. Due to ethical considerations, participants were asked to use an unidentified test account to log in to the RA system, but the final recommended portfolio was based on their inputting data. Their information would not be identified by the bank, and researchers erased their inputs manually after each test.

### 3.5. Data collection and analysis

This study collected three types of data: pre-survey data, a video recording of the interaction with the RA (AVG. = 19.8 min), an audio recording of post-interview (AVG. = 30 min), and a retrospective description is included in the post-interview. The interview recordings were transcribed into text and were imported into the software NVivo for analysis. The coding is analyzed inductively by finding themes with similar meanings. Three authors coded four interviews in parallel to validate the coding items and make the coding consistent within our research group. After the discussion, the initial coding metrics (an excel sheet with codes, sub-codes, and their definition) were confirmed, and

the rest of the interviews were coded by one of the authors. This author added codes inductively if new themes emerged in the following 20 interview transcriptions. The three authors synthesized and discussed these codes and annotations based on transcriptions.

In parallel with the interview transcription analysis, the video recordings were used as supportive data material when participants retrospectively discussed their behaviors and intentions when interacting with the RA. Also, each step of participants' interaction data with the RA system was transcribed into an Excel spreadsheet. These data included each participant's buttons, scrolling pages, documents being reviewed, and information filling in. For example, they checked “detailed information” of recommended funds/stocks, opened additional documents such as “factsheet”, reviewed “detailed cost”, and their input number indicating their investment willingness. This aims to determine if participants trust the AI-empowered recommendation by comparing the suggested amount of money and their actual input. This data can also reveal potential differences in patterns between participants with different levels of investment experience or financial understanding, namely, the expert and non-expert customers.

Because ten participants were not proficient with the language used in the RA, an automated translation (i.e., Google Translate) tool was used to help them perform the tasks. The translation accuracy was also checked by bilingual participants, proving to work well for conveying textual information. Due to the time lag for auto-translation, the time they spent was not accounted for in the data analysis.

The data from pilot tests ( $N = 2$ ) were excluded from the analysis. Thus, the data of 24 participants (14 females, 10 males; median age = 31.5 years) were brought into the analysis phase.

## 4. Results and findings

This section shows that the participants' perception of RA does not match their expectations. Non-transparency and insufficient information about the system make it difficult for participants to adopt the results generated by RA. However, data visualization showed its promise to convey comprehensible information to participants. The findings are synthesized from interview transcriptions and video recording analysis.

### 4.1. The perceived social role of RAs lags behind the participants' expectation

#### 4.1.1. Participants do not think they get advice when using the RA

As mentioned before, RAs are developed and implemented by service providers to partially replace human-resource costs. RAs also promise to provide the same quality of human-advisory service to customers with relatively low service and transaction fees. However, participants did not recognize that they were provided professional advice such as financial planning or wealth management. On the contrary, they felt forced to complete the purchase of certain financial products from the bank. The RA was perceived as a sales-centric service by the bank. P20: “... the bank wants to make a product that is easy for selling and make it easy for people to click the contract that they want me to sign.”; P18: “I was forced

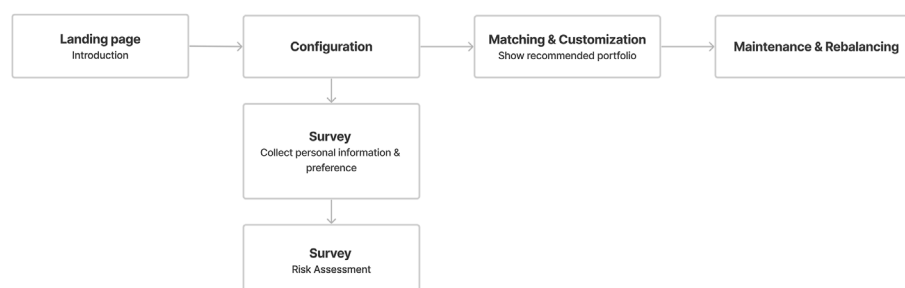


Fig. 1. The usage flow of general RA systems based on Nueesch et al. (2014).

into a fund.”; P7: “They do not recommend me anything, and this is how they make money.” Only two (P15 & P8) said they got suggestions and advice when using the RA. For example, P5: “RA helped me understand what to consider when planning my investment by answering different questions”. The reasons why participants do not perceive that the RA gives them advice are two-fold. First, the matching phase, in which algorithms run complex calculations, is not transparent. After answering the configuration questions, participants are shown directly to the final recommendation, and they are not informed about the process and criteria for picking up these financial products as their portfolios. Second, there is no opportunity for the participants to ask questions about the final recommendations; they must interpret and understand the portfolio by themselves. Therefore, the participants do not perceive that the RA gives advice. Instead, it barely sells customers financial products.

#### 4.1.2. *Participants do not think they are interacting with a “robot.”*

In the post-test interview, participants reported that they did not think the RA had any “robotic” features, and participants did not even perceive some significant features. The expectations they got from the name “Robo-” and the introductory information about the RA service shown on the onboarding pages did not match their experience with the system when using it. For example, some thought the service was a simple matching system: “very little robotic (...) a simple recommendation system (P20)”; it “just tries to match my profile with existing products (P21)”. Participants thought it was not intelligent because of the lack of conversational interactions like one expert with a customer: “I could not talk to, wouldn’t say more to the robots (P21)”; “the questions it asked me are very standardized, I don’t think it is my advisor. I would like my advisor to show more professionalism than this (P17).”

Participants also thought the service was “not intelligent” because it just stopped and did not give any information to help users recover from errors when they input unexpected or contradictory answers to the questions in the risk assessment phase. Three participants (P3, P11, P13) selected contradictory options when answering risk-related questions according to the analysis of video recordings. As per the regulations in financial services, customers must correctly answer these questions by themselves before they are allowed to proceed to the next step. Thus, RA systems cannot tell customers the correct one. However, in this test, participants were stopped from using the system. When this “contradictory” was detected, the service just “gave up” and asked participants to call the bank’s customer service. This unsatisfactory experience made participants feel that the system was not advanced. P13: “It (is) not the way that humans talk about finance. (...) It is unintelligent because it could have dynamically told me which answers are inconsistent.” P3: “I don’t know where I am wrong, and I don’t know how to fix it.” P11 went back to the previous stage and did the risk assessment again to proceed with this service and said, “In real life, I will quit the service, and I will not call them.” This unpleasant experience also made participants feel ashamed and caused dissatisfaction. P13 thought the system blamed him: “It told me that I did something wrong”.

Moreover, the features regarding automated maintenance and rebalancing of investment were not perceived by participants, even though this information was shown on the introduction webpage. It might be that the participants did not pay attention to the after-sale service because the test moderator asked them to terminate the test session before they had to perform the real purchase actions. However, after the researcher explained to participants that the system would perform automated maintenance and rebalancing after the investment, most participants said that they had not recognized this function of the RA when using the system.

#### 4.1.3. *Participants do not think a social robot can solve their problems but expect more human intervention*

In the interview, participants were asked if they thought that a virtual customer service robot or chatbot would be able to give more support compared to what the RA did that they just used. Almost all answered no

because they had not had any good experience talking with virtual customer services (P4, P6, P9; P26); based on their understanding of current technology, they thought that chatbots could only answer certain questions, and they preferred checking FAQ by themselves (P18, P24). Most importantly, P5 and P17 thought that it did not matter if humans or robots provided the service; what they cared about was that the service was performed professionally and rigorously and could be perceived as trustworthy.

Participants thought this automated service could be “pre-stage” before they went for human counseling because they still needed guidance and tutorial from a human advisor (P13). P16 thought a professional human could help to mitigate “the risk of miscommunication”, which might be caused by their misinterpreting the configuration questions from the system. Besides, P24 preferred to “have a combination between some person you can talk to and this automatic service” because he thought that talking to someone could make banks “know more about my economy, my goals and how I should act when I need an investment or loan”. P17 proposed that the bank should set a threshold for when a human advisor must do the final review: “For big investments that exceed a certain amount of money, a human advisor must get involved” because “investment can be a big decision, I cannot give my money to something based on a few clicks.” But P17 also said she might try this service by starting with a small amount of money. Thus, human involvement is expected by participants for different reasons, but mainly for calibration of the possible “miscommunication” and more personalized advisory suggestions.

#### 4.2. *Non-transparency is the main concern for decision making*

Non-transparency indicates the information required by participants that they do not think they get when using the system. This includes undisclosed information by the service and disclosed information but unperceived by the participant when using the service. Non-transparency mainly exists in two stages: risk assessment and final recommendation.

##### 4.2.1. *Non-transparency problems in configuration questions*

In the configuration phase, RAs are supposed to “understand” customers regarding their personal preferences, financial situation, and risk appetites based on surveys and answers filled by participants. This phase is fundamental because it not only gets necessary inputs from customers for the system to make complex computing; it also helps customers understand which factors construct the final decision (i.e., investment portfolio) made by this expert system. However, as the reasoning process is not adequately revealed, participants know neither the differences between each option, in particular, “high”, “minor”, and “intermediate”, nor where these answers will lead them to. This is not a survey that recommends any TV program but will lead to a high-stake decision on their real money.

4.2.1.1. *Participants want to know the logic behind the profiling questions.* The reasoning behind configuration questions and profiling strategies is unknown to participants. In the configuration phase, customers are required to provide information, so they will be categorized into different groups based on the risk assessment and other customized information, namely, overall savings, the income left after their compulsory monthly expenditure, loan, mortgage, etc. The standardized questions for categorizing the customers into risk-appetite groups are usually multiple-choice questions. Human advisors will help interpret and explain these questions and options and check for accuracy by regularly examining if customers understand them. In this user test, participants were much more aware and cautious of the logic behind these questions because they understood that these questions were to profile and match them with the appropriate financial recommendation. Thus, many participants were concerned about the issue of non-transparency because they asked why the RA asked these questions but

excluded some other questions that participants thought were more relevant to their financial planning (P16 & P18). Participants were unsure how their answers resulted in the final recommendation in this system and to what extent the different answers would result in different investment portfolios. P7: “If I chose a bigger number. Would they recommend something different?”. The reasoning that matches participants’ answers to the risk-appetite groups remains unknown and unpublished. Is it just a risk assessment the same as a human-advisory service, or does the algorithm play any role here? No explanation is offered.

**4.2.1.2. Participants have difficulty and uncertainty when answering questions.** Without guidance or reference, it was difficult for participants to accurately interpret or comprehend each question and its corresponding options. This uncertainty causes distrust of the results generated by those inputs. When answering multiple-choice questions, participants felt uncertain when having to interpret different options and select one because of vague terminologies such as “minor”, “intermediate”, and “high”. Especially when they were asked questions regarding risk preference. Most importantly, they were concerned about whether an inaccurate answer or input would lead to a recommendation that did not fit them. Participants asked for more comprehensible and concrete definitions. P5: “It is unclear what is a ‘high’ risk here because it is subjective for different people. It should also define that.” P3: “They didn’t give me concrete numbers or quantified it in a good way. How much money is minor? 1000 out of 10,000? Or 1 out of 10,000? I am afraid people have different understandings about ‘minor’.”

This lack of transparency at the early stages unsatisfied participants and distrusted the results based on incorrect inputs. The relationship between the customer and the service that the participants expected was perceived to be broken. Participants were skeptical about how valid the final recommendation was if it was completely based on how well they understood the questions, which might not have been accurate enough. P5: “When answering, I just select one with feeling. So my answer is not that valid. (...) I cannot trust the result if I cannot provide solid information that I know exactly what I want.” P12: “Because when I put medium risks, I was unsure between the low and the medium.” P3: “If I adopt a product (portfolio) that doesn’t suit me because I didn’t understand the survey question correctly, I might bear more risk.” This uncertainty and distrust caused by the non-transparency at the very early stage will persist throughout the user experience and affect the following interactions.

#### 4.2.2. Transparency problems in the final recommended portfolio

After configuration, the algorithm will automatically match the customer’s profile to a certain group and show them the final recommendation, which is usually a portfolio of ETFs (exchange-traded funds) that is a basket of investments such as assets, stocks, bonds, etc. This is a passive investment and is not fully transparent because the manager’s trading strategies are normally unpublished. However, in the test, most RA participants were unaware of the portfolio and how ETFs work in the financial market. Based on the information provided by RA, participants’ confusion is two folds: lack of information about the mechanism in the system that creates the suggested portfolio; and the lack of information that they think is necessary for their decision-making.

**4.2.2.1. Lack of transparency of the recommendation system mechanism.** - After reviewing the recommended portfolio, most participants asked for the result’s source and principles. This non-transparency revolves around the AI-empowered algorithm and its mechanism for advising customers on the optimal and personalized portfolio. P9: “I don’t know how and what the criteria are for picking up the stocks for me.” P10: “I want to know more specifically where these numbers suggest my monthly deposit comes from.” This lack of transparency negatively affects participants’ trust as the service claims that the recommendation is “tailored” for them

and “adopted” to their needs; however, participants did not perceive those features. P18: “I do not understand what kind of models or mechanism the recommendation is based on.”

The participants were aware that they were categorized into different risk groups, and they would like to know the final recommendation in other groups, even though the participants did not explicitly explain the reasons for this requirement. P20: “why cannot I review all (plans) choices at the end of the process? And that is also a factor where I would lose trust in this service.” P17: “I know they might have seven groups in total. I was classified into five; I want to see what the products in other groups are.” P5 makes an analogy with personality tests: “This could be, okay, our (banks’) options are this and this, just like the personality test, you know. Then you can see other people (’s distribution) as well.” This finding is in line with previous work regarding AI’s use in the medical domain (Aoki, 2021); humans show more trust in AI’s decisions if they feel that they are still involved in the decision-making by communicating with the automated system. This type of “communication” is lacking in RA service.

**4.2.2.2. More transparent managerial strategies are needed regarding the components of the recommended portfolio.** Participants tended to show more consciousness as they requested more detailed information about the companies of bonds and securities, which were about the investment targets in the portfolios. P13: “So 22% of the investment will go into the IT sector, and 13% here will go into the health (industry), but it doesn’t show me what’s the company?” Those participants who selected “sustainable preference” were more ethically concerned investors, and they thought it was important for their decision-making to know the criteria for screening sustainable investments and what companies the bank would invest in. P12: “I’m surprised it’s (sustainable) going; the majority is going to IT. I didn’t expect that. Like, I thought it would go more towards energy and sustainable energies.” P21: “Because if I’m an ethical investor, I want that information to be transparent. If I know what they are, I can track the companies and see what they are doing.”

#### 4.2.3. Some information is transparent but not easily navigated

In financial service encounters, information disclosure is strictly regulated by law. Customers can casually ask questions they most care about to human advisors, and customers get informed in this conversational-based process. However, when interacting with RAs, participants should be more self-reliant. Even though the system discloses enough information, some are deeply hidden. For instance, some PDF documents are not integrated with the user interface but requests customers to click a new webpage. Customers are requested to have a high capability to search for information and actively navigate themselves, which is overwhelming and impossible for those non-expert users. According to our observations of video recordings, expert participants can easily navigate information to help understand the performance of the recommended portfolio or the service terms. For example, P4, P20, and P26 actively open documents such as fact sheets to review past performance and ratings about each specific portfolio component.

In the interviews, participants were concerned that they were not informed about the maintenance and quitting terms if they would invest money using this system. Specifically, after being informed that the system would automatically rebalance their portfolio, participants wondered how much information would be reconciled with them; and when they would get a notification that there would be an auto rebalancing. P13: “The service contract that I signed today might not include the algorithm to be dynamically changed in the future. And these changes are not published nor public.” P9: “I don’t know if I will be regularly pushed or updated with changes after my investment, such as the trend, growth, or loss. But now I don’t know and don’t get any information if I can receive these notifications. I want to follow and track my investment.” Participants were also concerned about how much they would lose if they dropped out earlier than the agreed plan (e.g., P8 &

P11). Some noticed a text saying customers can take their investment back whenever they want. But they thought the information here remained simple and unclear (e.g., P6). Participants expected the RA can prioritized information as their importance, in particular, the possible costs and service fees, because customers of financial services indeed care about and are sensitive to them.

#### 4.3. Insufficient information and the lack of domain knowledge for adopting RA's results

Insufficient information is the second issue that hinders participants' adoption. Generally, non-expert customers with limited investment experience or financial understanding need expert systems like RAs to make financial decisions. In the interview, when asked, "would you adopt the recommendation or use RAs in the future" most participants were uncertain and could not make decisions based on their experience with RAs. Only four gave certain answers like "yes" (P1 & P14) and "no" (P20 & P24) after the test session. Moreover, when being asked "if you want to use and try it several times and then make decisions", expert participants were more confident regarding their first trial and believed that the result would be the same if they tried the whole procedure again (e.g., P4, P9, P17, P20, P24). At the same time, non-expert participants showed more uncertainty and distrust of the decision made by the system. Non-experts need more guidance in providing access to information that is transparent but also explanative and understandable for them.

##### 4.3.1. Non-expert participants require more understandable information to support their adoption decision

Participants said they needed more information to understand the final recommendation and to support their decision-making. The information is two-fold: information disclosure regarding the RA service; and financial understanding, which is basic domain knowledge to support interpretations and comprehensions of those disclosures. Differences exist between expert and non-expert participants. The latter felt overwhelmed when so much information appeared at the last stage. Even though some can find these documents, it was still hard for them to understand the content, including financial terms and numbers, not to mention how to utilize this information to support their decision-making. Generally, non-expert participants expect to learn basic financial market and investment knowledge through words they can easily comprehend. P10: "I open the fact sheet and realize; I don't understand the terms." P12: "I expected it could tell me what the stock market is like, such as the current index of the whole stock market. I hope that someone can guide me with more basic information." P6: "They should have a good editor to create understandable texts. Or they have links to some knowledge where I could learn more." In short, extra guidance in the system is expected to reach three aims: (1) to help customers navigate the information they need; (2) to guide them to understand this information, and (3) to tell them which are the most important criteria for evaluating and predicting the performance, of a fund for example.

##### 4.3.2. Participants expect RAs to have a holistic understanding of their personal information

Participants were skeptical that the RA could provide a holistic and personalized portfolio based on the limited information currently collected from them through the system. They also expected to spend a long time using this service rather than getting a plan within a few minutes. P17: "Before using (the RA), I thought it would ask me more questions that allow me to see the value of this service. These questions are so standardized I don't think they can give me good suggestions on an individual level. I doubt it." P7: "It is a short (the time), honestly. It can be more patient with me. It's a big decision. It's not something I would like to have done in 3 min"

Thus, some participants would like that the system could understand more about users by asking more personalized questions such as

everyday finance and any long- or short-term plans in an individual's real-life scenarios, for instance, pensions (P21), buying a car or an apartment (P8), or other plans in the future. P21 thought the RA skipped important questions about customers' horizons and intentions for saving and investing. This also makes them feel that the advisor is not personalized as they see from the introduction page, and this unmet expectation negatively affects their experience.

##### 4.3.3. Data visualization is perceived as an effective tool to convey comprehensible information

In the interview, most participants gave positive feedback in terms of the interactive data visualization in the RA. Data visualization instruments show great potential in three aspects: visualizing financial understanding (especially risk assessment), interpreting the performance of the recommended portfolio, and envisioning growth and loss.

In the selected RA, two types of data visualizations are utilized. One visualizes risk assessment by showing concrete numbers in discriminative visual cues (i.e., two colors) (Fig. 2). The other visualization presents more information and interactive parameters by which customers can envision the growth and loss of the money they plan to invest in one-time or monthly deposits (Fig. 2).

Compared with the vague terminology in the option such as "low", "middle", and "high" in risk assessment mentioned before in this paper, the data visualization (Fig. 2 Left panel) is claimed to be "more concrete (P12)" and "much more tangible (P5)" in helping participants understand risk, even though some design problems exist (e.g., most participants did not realize the bottom round button is interactable). All participants said that the interactive line diagram (Fig. 2 Right panel) provided sufficient information for them to understand the performance of their portfolio and how the possible growth and loss of their money could change over time. First, the interactive line diagram is self-explanatory and comprehensible for participants without much knowledge about this type of service. P10: "it can provide me the estimation of the lowest and highest (...) It can give you a general conception, like the rate, percentage. Then I know the highest money loss I can bear, concretely and practically." P21: "This (Fig. 2 Right panel) is more comprehensible regarding the time horizon and related risk". From the video analysis, every participant interacted with the tool by inputting different variables in Fig. 2. P4 thought the input in time horizon helped to associate risk with the concrete amount of money, making the risk more understandable.

##### 4.4. An untransparent system and incomprehensible information: where the distrust comes from

Participants showed distrust and suspicion of the result if the RA system was not transparent or incomprehensible to them. According to the interviews, participants' decision to adopt the RA was affected by the system and other factors, such as unsatisfied fees and initial intention of using RA. However, for both expert and non-expert participants, transparency and sufficient comprehensible information are crucial for their willingness to use RA or not. Besides, it is worth noting that most participants distrust the system from their interviews and the screen recording. After reviewing the final recommended portfolio, participants were asked to input their investment plan to finish the task performance. In the video recordings, most participants did not follow the RA's suggested amount of money. Only four (P6, P7, P11, P18) inputs exactly matched the amount of money the RA suggested. This can be interpreted as participants not taking the RA's suggestion as their best fit, thus representing distrust of the risk grouping and the mechanism behind the recommended portfolio. Most participants showed that they trusted the bank and the technology, being confident that there would be no fraud. However, their experience of the whole system makes it difficult to trust the result (P3, P5, P6, P7, P12, P13, P21). According to the interview data, this distrust negatively affects participants' willingness to adopt the advice generated by the RA.



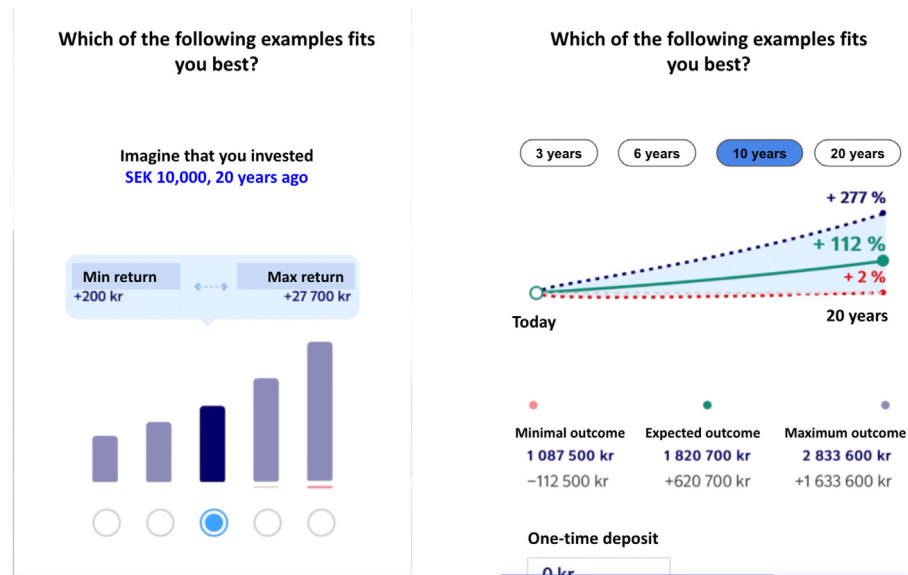


Fig. 2. Left panel: The data visualization in the risk assessment phase. Right panel: the interactive diagram in the recommendation portfolio. These are not the original interfaces for translation, but the contents are the same.

## 5. Discussion

This holistic user-centered study of a functional retail bank RA aims to understand how the AI-empowered system can impact customers' perception and adoption. Methodologically, the approach of asking customers to use a functional RA compensated for the previous empirical studies by simply conducting surveys without considering customers' experience of interacting with a RA in real-life situations. Theoretically, this study extends the knowledge of RA perception and adoption research by articulating the negative effects of the lack of transparency and comprehensibility in a RA system. In particular, this paper claims that non-expert customers request more comprehensible domain knowledge to use and adopt RAs. Besides, the perception of RA lags behind both customers' expectations and the service providers' ambitions. Overall, the significance and the role of RA's system design have been levered up, and some design takeaways, current limitations, and future work are also discussed.

### 5.1. A transparent and comprehensible AI-empowered system is significant for RA perception and adoption

In the context of financial investment, transparency and comprehensibility are important because the investment decision is relatively high-stake, and investors are "vulnerable customers" (Beltramini, 2018; Mogaji et al., 2021), which means their perception of RA can be negatively affected by lack of transparency. They are not resilient or tolerant of unmet expectations. Moreover, unlike other AI systems, investment is not a "one-time" behavior, but a long-term and dynamic process. The whole service provided by RA ranges from portfolio design and maintenance, which is difficult for customers to understand. Customers are highly dependent on written texts and visualized data without human intervention when using RA systems. This also increases the threshold to adopt RA. Thus, a good RA system should be transparent and comprehensible.

By extending prior studies in the web-based Fin-tech (Kilic et al., 2015; Nueesch et al., 2014, 2016; Nussbaumer, Matter, Reto à; Nussbaumer & Matter, 2011; Nussbaumer, Matter, & Schwabe, 2012; Porta, et al., 2012), the result has levered up the role and importance of a transparent and comprehensible system perceived by customers, which works together with other multiple factors affecting customers' perception and adoption of RAs. The mechanism behind consumers' behavioral

intention is complex, especially when AI meets with financial decisions. This interdisciplinary paper bridges the previous fragmented studies in different fields (Belanche, Casalo, & Flavián, 2019; Ben David et al., 2021; Ben-David & Sade, 2020; Cheng et al., 2019; Gan et al., 2021; Shin, 2021a; Wu & Gao, 2021), by compensating for the lack of customers' real use of a functional RA in previous studies.

#### 5.1.1. Transparency issues running through the whole system

Transparency has long been an issue in financial advisory services (Kilic et al., 2015; Nueesch et al., 2014, 2016; Nussbaumer & Matter, 2011) and AI-empowered customer experience design (Ben David et al., 2021; Shin, 2021a). This empirical study also revealed different transparency issues embodied in AI-empowered advisory and financial systems. Customers in financial service encounters are more aware of issues because the investment decision is high-stake compared with other AI-empowered services. This study articulates these issues, and it indicates that transparency should be a principle that runs through every detail of a RA system ranging from the texts on the introductory page, the reasoning behind survey questions and options, the mechanism of the recommended portfolio, the performance and components of financial products, the service fees, to the service-related terms, such as termination. Customers using RA are more rational than those lay investors in conventional financial services with the popularization of AI-empowered technology and its wide use. They would like to make more informed decisions even though they trust the service providers and the technology, namely the algorithm behind the system. Similar to the finding of previous internet-bank research (Nussbaumer, Matter, & Schwabe, 2012), transparency in the AI era intensively requests the system not only provide the necessary information, but also prioritize the importance of different information and convey these to customers. This can help to balance the information asymmetry between an AI-empowered expert system and customers in financial service encounters. Otherwise, the non-transparency will hamper the relationship established on the trustworthiness between customers and service providers.

#### 5.1.2. Augmenting financial understanding: comprehensible transparency

Previous research shows customers with a low level of financial literacy are inclined to use RAs (Brenner & Meyll, 2020), and they need an AI-empowered expert system more urgently to give recommendations than expert customers. Also, non-expert customers with little knowledge might feel embarrassed and anxious when conversing with a human

advisor (Gerrans & Hershey, 2017); the automated service is supposed to be a better alternative. However, it was found in this study that most non-expert participants did not get sufficient support when using the RA. First, they cannot navigate themselves through this overwhelmingly informative system. Second, they cannot correctly understand financial terms, and the lack of domain knowledge causes distrust of the AI-generated result. This finding is in line with a prior study that claimed that the lack of basic financial knowledge could lead to distrust of the result generated by AI, even though they are accurate (Dikmen & Burns, 2022). At the same time, customers without a basic understanding of how the financial market would be or what risk they will take would be put in danger by blindly adopting RAs' recommendations. In our study, some non-expert customers with a master's degree had difficulty correctly understanding terms like "interest" and three participants selected conflict answers when doing the risk assessment. Thus, those non-expert customers expected that RAs should provide them with transparency and understandable information that could support their financial decision-making. In the human advisory service encounter, advisors can check if the customers understand the questions correctly. Human advisors can also use different pedagogical tools to address customers' lack of financial literacy, enhancing the customers' understanding of personal finance and the effect of different investment choices.

#### 5.1.3. Potential of interactive data visualization

The interactive data visualization in the RA service has proven its potential to help customers understand risk, envision the growth and loss of their assets, and associate information with the time horizon. By using different colors as discriminative visual cues to indicate different situations, this visualized design synthesizes parameters from different sources (e.g., AI-driven models, input from customers), then embodies the complex mechanism into an interactive diagram that can easily fit with customers' cognitive ability. Thus, this user study can treat interactive data visualization as an explainable AI approach. Moreover, this design shows the recommended deposit plan and still grants customers autonomy to input and adjust their money to be invested. This finding extends previous research regarding interactive data visualization in making judgments (Eberhard, 2021), understanding information (Perdana et al., 2018), and financial decision-making (Tang et al., 2014) by uncovering how well it is used and its potential in automated financial advisory contexts.

#### 5.2. The discrepancy between RAs' social design and customers' perception

It is a disruptive finding that customers' perception of the RA service is not in line with the intentions and ambitions of the service providers. Findings in this study have extended the knowledge regarding RAs' weaknesses (Jung, Dörner, Weinhardt, & Pusmaz, 2018; Jung et al., 2019). Most importantly, these findings have questioned in a disruptive way the assumption that RAs' anthropomorphism can compensate for the lack of human intervention in RAs (Adam et al., 2020; Deng & Chau, 2021; Hildebrand & Bergner, 2021). According to our user study, customers need a human advisor to double-check their comprehension of aspects of the service, primarily due to the possibility of "miscommunication" in the configuration phase. Here, a more transparent and understandable information-sharing mechanism between customers and the system regarding the interpretation of the questions and each option connected to them is lacking. Without knowing why these questions are asked and what their answer can lead them to, customers need and expect a human advisor as a bridge to confirm if customers understand the question the same way as was intended by the developers of the system; and if the system interprets participants input in exactly the way that the participants meant. Different levels of human-likeness design might affect consumers' behavior and decision-making in certain ways; however, based on the findings in this study, that is not what customers need nor expect.

The perceived performance of the RA in markets studied in this test has lagged behind the visions of their developers. The service provider of the RA investigated in this study claims that their RA service does not compromise on the service quality but equals a human-advisory service. However, most participants do not recognize that they get valuable advice or suggestions for their personalized wealth planning, which is an essential part of human-advisory service. Second, customers who are inexperienced in investment or have a low financial understanding do not get sufficient information or knowledge by using the RA service to support their decision-making. Even worse, these customers felt "blamed" and "abandoned" by the system when they input answers that the algorithm could not compute because these were contradictory to what was accepted by the system configuration.

#### 5.3. The dilemma between system design and business implementation: practical takeaways

Here we propose takeaways for the practitioners and service providers of RA. In practice, the implementation of RAs is not simply user-centered due to multiple stakeholders' involvement, because RA developers and designs are only part of the business strategies. In behavioral finance studies, customers' financial decisions tend to be biased (Chira, Adams, & Thornton, 2008; Costa, de Melo Carvalho, de Melo Moreira, & do Prado, 2017). Some customers would only invest in those with high returns and ignore other factors, and lay investors with limited financial literacy would not actively acquire any information but still make investment decisions. If conventional financial advisory service is sales-centered, then RAs are supposed to be result-driven. It is designed to encourage customers to start in a very small amount by showing them the results quickly, so customers can decide based on the return rate or service fee. Thus, it is understandable that existing RAs seems to be easy to use and their "smooth" and "simple" interface makes customer rapidly check the final result because they do not want to let customers leave halfway without seeing the financial products, which are their core selling points of the service.

According to the findings of this study, customers are more self-aware and patient when interacting with AI-empowered systems in making financial decisions. They are willing to spend more time and get sufficient information from RA systems, but service providers underestimate this. The system design is supposed to be more user-centered, and the decision mechanism and reasoning process should be transparent and understandable for both expert and non-expert customers. Most importantly, data visualization can be further explored to convey sufficient and explainable information as an effective instrument for the interface.

#### 5.4. Limitations and future studies

This study did not employ standardized assessments to evaluate participants' financial literacy. Instead, this information is incorporated in the pre-test survey, and participants were asked to self-report financial knowledge on a 1 to 5 scale (scale 1 stands for very little knowledge; 5 stands for expertise knowledge). Possible biases exist in participants who are overconfident or underestimate their financial knowledge. Further work regarding the comparison between user groups with different financial literacy would be a very interesting avenue. Also, future work could be extended from focusing on behavioral intention to including long-term outcomes, for example, customers' retention or after-adoption behaviors.

### 6. Conclusion

This study conducts a user study regarding a fully functional AI-empowered automated financial advisory system and its provided service. The findings show that the social aspects that are supposed to be provided by the RA are not perceived as such by participants. The lack of

transparency and incomprehensible information negatively affect participants' trust in the system and makes it difficult to adopt the results generated by RAs. This research extends previous research by investigating a RA and understanding customers in an in-depth empirical user-centered study in a real-life situation. It contributes to AI-empowered system design in the financial service context and provides an understanding of customers in automated service.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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